

Seent: Interfacing Gamified Olfactory Training

Ling Qin
Imperial College London & Royal College of Art
London, UK
ling.qin@network.rca.ac.uk

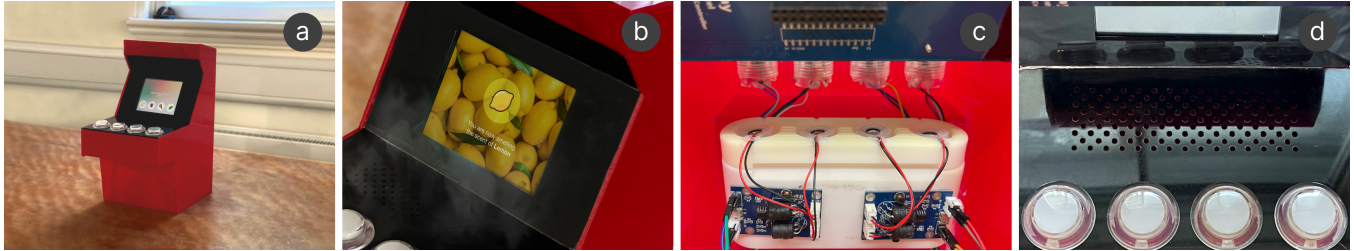


Figure 1: a) Seent is an interactive interface designed to gamify olfactory training with an arcade-style appearance. b) The Graphical User Interface (GUI) of Seent. c) Inside Seent is an array of piezoelectric atomisers d) The control panel of Seent.

ABSTRACT

Olfactory dysfunction affects a significant proportion of the global population. Although traditional olfactory training can help with rehabilitation, it often lacks engagement due to its repetitive and dull nature. Advances in digital olfactory technologies have made the interaction between humans and odours possible. We introduce Seent, an interactive interface designed to gamify olfactory training, including hardware to digitalise odours and a corresponding Graphical User Interface. Seent integrates playful gameplay into the training process, allowing participants to engage in a series of odour-based interactive games. This paper concludes by outlining future steps and proposing its potential to enhance olfactory rehabilitation and promote regular olfactory health monitoring.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

Smell; Human-Computer Interaction; Olfactory Interface; Multi-sensory Interfaces; Digital Health; Well-being; Interaction Design

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1 INTRODUCTION

It is estimated that around 20% of the worldwide population suffer from the loss of olfaction [5, 14, 23]. The prevalence rises to 62% for those who have suffered from the global pandemic since 2019 [18]. This recent health crisis has also raised unprecedented awareness about the critical role of the olfactory system in human well-being. Olfactory dysfunction can severely decrease the quality of life [11], affecting safety perceptions such as detecting fire, gas, and smoke [13], reducing enjoyment of food [3], impairing the ability to detect spoiled or contaminated food [3, 21], increasing concerns about personal hygiene [3], and even leading to social dysfunction [21].

The loss of olfaction is not irreversible. Regular olfactory training can assist individuals with olfactory dysfunction in the rehabilitation of their olfactory function [10, 15, 17]. However, such training is highly repetitive and dull, resulting in low engagement. Previous research has demonstrated an interactive olfactory training solution in which odours are delivered through a miniature air pump with a companion app [1]. Additionally, cross-modal of visual and olfactory stimulus may contribute to better training results [19]. The absence of playfulness remains in these solutions.

In this work, we introduce Seent, an interface enabling participants to interact with a series of olfactory games through the digitalisation of odours and a corresponding GUI. These games start with a 'press and play' mechanism to expose participants to various odours, followed by an odour guessing game and an odour differentiation game. This gamified interactive olfactory training process automates the conventional method while making it more playful.

2 OLFATORY TRAINING

Traditionally, olfactory training has been highly manual and repetitive. It is conducted by exposing individuals with olfactory dysfunction to 4 different odours, typically phenylethyl alcohol (rose odour), eucalyptol (eucalyptus odour), citronella (lemon odour), and eugenol (clove odour). Participants inhale each odour from sniff

jars or bottles in a random order for 10 seconds, twice a day, over a period of 12 weeks [10, 11].

3 SEENT

Gamified cognitive training has been shown to be more engaging and increase the motivation of participants, potentially improving their performance [12, 16]. To automate the conventional training process while enhancing engagement and playfulness, we developed Seent, a gamified interface for interactive olfactory training. The hardware delivers specific odours and integrates with a GUI that visualises these odours and the gaming process, allowing participants to interact with the system. Four buttons on the panel are used for program control. A 5-inch screen with a resolution of 800*480 is used to display the user interface.

3.1 Digitalisation of odours

The digitalisation is based on an array of four piezoelectric atomisers attached to the cartridges, as shown in Figure 1c. Each cartridge holds a 4% solution of phenylethyl alcohol, eucalyptol, citronella, and eugenol respectively, similar to the conventional method. The piezoelectric atomiser comprises a metal microporous plate bonded to a circular piezoelectric ceramic plate. When an electrical signal activates the ceramic plate, it vibrates at 2.4 MHz, causing the plate to atomise the solution into a mist without heating or chemicals. A cotton bar soaked in the solution contacts the atomiser, facilitating the mist generation and delivery through the microporous plate. The mist escapes through holes on the panel (Figure 1d), allowing participants to sniff it. The system is driven by a Raspberry Pi 4B.

The primary advantage of digitalised odour delivery is its precision and consistency, which are lacking in conventional training methods [2, 9]. Seent dispenses a uniform odour volume at controlled intervals. This alleviates the need for participants to manually time their sessions, as is required with conventional methods where they must hold jars in one hand and simultaneously keep track of time. This automated process also enables the integration of gamification into the training process.

3.2 Gamified olfactory training

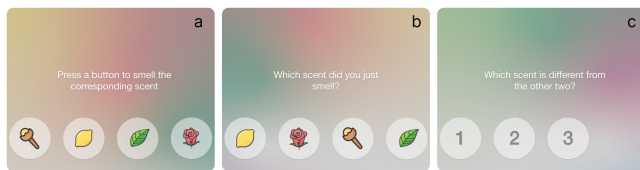


Figure 2: a) A press & play game. b) An odour guessing game. c) An odour differentiation game.

The gamification is achieved through a series of odour learning, odour guessing, and odour differentiation games. The learning process (Figure 2a) begins with 'press and play'. Participants simply press any button, and the corresponding odour will be released for 10 seconds. Each button corresponds to an icon of the odour displayed on the screen. During the odour release, while the user's nose is focused on the odour, they can see what it is on the screen,

as shown in Figure 1b. This visual clue, including the icon and the imagery of the odour, has been proven to enhance odour perception [6, 20]. Participants can practise in any preferred sequence. During the second game session (Figure 2b), one of the four odours is dispensed at random, and participants are asked to identify the odour they just smelled by pressing one of the buttons on the panel. This game session contributes to the improvement of odour identification, the ability to recognise, identify, and name an odour correctly. The following game session focuses on odour discrimination (Figure 2c), the ability to detect and differentiate between different odours. For the odour differentiation game, participants are exposed to a series of three odour releases, in which two of the three releases have the same odour, while one is different. Participants are then tasked with identifying which release is different. For the latter two games, feedback is displayed immediately, indicating whether the response is correct or wrong. This allows participants to not only be exposed to the odours but also to keep track of their progress. In summary, Seent enhances playfulness by integrating game elements, concealing odour information to increase playability, and providing instant feedback.

4 CONCLUSION & FUTURE STEPS

We presented Seent, an interactive interface that digitalises odours and transforms the conventional olfactory training process into a series of playful games. Seent emphasises playfulness in olfactory training by introducing a controllable odour delivery system and an intuitive GUI, making the process more accessible, engaging, and enjoyable. We speculate that incorporating game elements and immediate feedback mechanisms can encourage participants to engage more consistently, thereby benefiting their olfaction rehabilitation. Moving forward, our next step involves inviting individuals with impaired olfaction to participate in user studies to evaluate this hypothesis. These studies aim to assess its impact on olfactory sensitivity and user experience compared to traditional training methods.

For further steps, we seek to expand the application of Seent from olfactory training to olfactory testing. Olfactory identification and olfactory differentiation tests are commonly used for clinical olfactory assessment [24]. Beşevli et al. recommended that individuals be encouraged to undergo regular olfactory checks, just as we routinely check our eyesight and hearing [1]. Self-monitoring and self-care of olfaction should be a part of our daily lives [1, 17]. Besides, Parkinson's Disease and Alzheimer's Disease come with a noticeable symptom of olfactory dysfunction [4, 7, 8]. Conducting olfactory function tests is economical and simple. These tests are potential clinical markers for the severity and progression of both diseases [22, 25]. We therefore propose Seent can not only be used for olfaction rehabilitation, but also for olfaction testing. We invite the human-computer interface community to explore and unleash the full potential of Seent, driving innovation and awareness in olfactory health.

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